# Quantum Computing

Quantum Search and Its Limits

Lecture 5 Notes

Ralph Razzouk  Marc Fares  Joseph Jabbour

Lebanese Science Organization

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## About these notes

These notes accompany a nine-lecture course on quantum computing designed and produced by the Lebanese Science Organization. Each chapter corresponds to one lecture. The lectures are concept-focused: they explain what a quantum state is, how entanglement, interference, and measurement combine to produce computation, and what is required to make that computation reliable. Linear algebra, complex numbers, and elementary probability are assumed.

Each chapter names its lecturer and date, opens with a short preface, develops its material with the derivations in place, and closes with a summary. Worked examples appear in numbered boxes, as do definitions, theorems, remarks, and common misconceptions. Works cited in the text are collected in a single reference list at the end of the document.

Conventions used throughout: qubits in a register are labelled from left to right, so that in $|ab\rangle$ the first qubit is $a$; the first qubit is the top wire in every circuit diagram; the computational basis of two qubits is ordered $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; circuits are read from left to right, so a gate $U_1$ followed by a gate $U_2$ acts as the matrix product $U_2U_1$.

This chapter is exported individually from the full lecture notes (5 of 9). These notes are made available for open, public study. Reuse conditions will be stated separately by the Lebanese Science Organization.

# Contents

**1 Quantum Search and Its Limits** **3**
1.1 Unstructured search . . . 3
1.1.1 The phase oracle . . . 4
1.1.2 The single-query attempt . . . 4
1.2 The Grover iteration . . . 5
1.2.1 Inversion about the mean . . . 5
1.2.2 Rotation in the invariant plane . . . 6
1.2.3 The four-item case . . . 7
1.2.4 The iteration circuit . . . 8
1.3 Query complexity . . . 8
1.3.1 The optimal number of iterations . . . 8
1.3.2 Several marked items . . . 9
1.3.3 The lower bound . . . 9
1.3.4 Limits of the speedup . . . 10
1.4 Summary . . . 11

**References** **12**

# Chapter 1

# Quantum Search and Its Limits

**Lecturer:** Joseph Jabbour

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We are given a function that recognizes a solution when shown one, and nothing else. This lecture asks how interference can locate a solution among $N$ candidates faster than examining them one at a time, and how much faster it is possible to go. The mechanism is amplitude amplification: the Grover iteration composes two reflections into a rotation that steadily increases the amplitude of the marked item, so that about $\sqrt{N}$ queries suffice where a classical search needs about $N$. We derive the optimal number of iterations, treat the case of several solutions, and see that the quadratic speedup is the best any quantum algorithm can achieve for unstructured search.

## 1.1 Unstructured search

Let $f\colon \{0,1\}^n \to \{0,1\}$ be a function on $n$-bit strings, and write $N = 2^n$ for the number of inputs. An input $x$ with $f(x) = 1$ is called marked, or a solution. The set of marked inputs is $W = \{x : f(x) = 1\}$ and we write $M = |W|$ for its size. For most of this chapter we take $M = 1$ and call the unique solution $w$; the general case is treated at the end.

**Definition 1.1: Unstructured search**

The unstructured search problem is to find some $x$ with $f(x) = 1$, given access to $f$ only through queries: we may submit an input $x$ and learn the value $f(x)$, and we have no other information about $f$. The cost of an algorithm is the number of queries it makes.

The word unstructured is the operative word, and section 1.3.4 returns to it. It means that knowing $f(x)$ for some inputs tells us nothing about $f$ at other inputs. There is no ordering to exploit, no gradient to follow, and no algebraic relation among the values. A classical algorithm can therefore do no better than to query inputs until it finds a marked one. With one marked item among $N$ this takes about $N/2$ queries on average and $N - 1$ in the worst case, after which the last input is known without a query; with $M$ marked items it takes about $N/M$ queries on average.

More precisely, a classical algorithm in the query model learns one value $f(x)$ per query and, by the assumption of no structure, learns nothing about other values. After $k$ queries without success, the marked item is uniformly distributed over the $N - k$ unqueried inputs, so the probability of success on the next query is $1/(N - k)$, and the expected number of queries to find a single marked item is about $(N + 1)/2$, or exactly that if the last remaining input is also queried rather

than deduced. With $M$ marked items the expected number is about $(N+1)/(M+1)$. Grover's algorithm [1] replaces these by about $\frac{\pi}{4}\sqrt{N/M}$, and Theorem 1.2 says this is the best possible up to a constant.

Two tasks must be kept apart. Recognizing a solution means evaluating $f$ at a given $x$; by assumption this is cheap, one query. Finding a solution means producing an $x$ with $f(x) = 1$; this is the expensive task, because a priori every one of the $N$ candidates is equally likely to be the one. The gap between the cost of recognizing and the cost of finding is exactly what the search problem measures, and it is the gap that Grover's algorithm narrows. It does not remove it.

### 1.1.1 The phase oracle

We reuse the oracle model of **??**. The function is available as a reversible gate $U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$, and the phase kickback argument of that chapter, with the output qubit prepared in $|-\rangle$, turns $U_f$ into an operator acting on the input register alone.

**Definition 1.2: Phase oracle for a marking function**

The phase oracle for $f$ is the unitary $O_f$ on $n$ qubits defined on the computational basis by

$$O_f |x\rangle = (-1)^{f(x)} |x\rangle . \tag{1.1}$$

It reverses the sign of the amplitude of every marked input and leaves the amplitude of every unmarked input unchanged. One application of $O_f$ counts as one query.

With a single marked item $w$ the oracle has the closed form $O_f = \mathbb{I} - 2|w\rangle\langle w|$: it acts as $-1$ on $|w\rangle$ and as $+1$ on every state orthogonal to $|w\rangle$. Nothing in $O_f$ reveals $w$ directly. Applied to a basis state it changes at most a global phase, which **??** showed to be unobservable. The oracle becomes informative only when it acts on a superposition, where the sign becomes a relative phase.

### 1.1.2 The single-query attempt

That observation suggests the following attempt. Prepare the uniform superposition of **??**,

$$|s\rangle = H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle , \tag{1.2}$$

and apply $O_f$ once. The state is now $|s\rangle - \frac{2}{\sqrt{N}} |w\rangle$, which does depend on $w$. But measuring it in the computational basis yields $w$ with probability $|-1/\sqrt{N}|^2 = 1/N$, exactly the probability of guessing $w$ at random. Measuring $|s\rangle$ itself, without any query, also yields $w$ with probability $1/N$. Preparing a superposition over all inputs is not a search algorithm. It puts every candidate into the state with equal weight, and the Born rule then returns a uniformly random candidate. The marked item is present in the superposition, but so is everything else, and the measurement has no reason to prefer it.

**Wrong Concept 1.1: A quantum computer tries all $N$ inputs at once**

The uniform superposition $|s\rangle$ does contain every input, and $O_f$ does act on all of them in one application. But there is no operation that reads off which component acquired the sign. A measurement samples one component according to the squared amplitudes, and after a single query every component still has squared amplitude $1/N$. The work of a quantum search algorithm is not the parallel evaluation, which is free, but the interference that moves probability onto the marked component afterwards. That work costs repeated queries, and

how many it costs is the subject of the rest of this chapter.

The sign flip produced by $O_f$ is a relative phase, and **??** taught that a relative phase becomes observable only after a change of basis makes the amplitudes interfere. The next section constructs the change of basis that does this for the search problem, and shows that it must be repeated.

## 1.2 The Grover iteration

Two vectors are distinguished: the marked state $|w\rangle$ and the uniform state $|s\rangle$. They are not orthogonal, since $\langle w|s\rangle = 1/\sqrt{N}$, and they span a two-dimensional real plane inside the $N$-dimensional state space. Everything that follows takes place in that plane. Let $|w_\perp\rangle$ be the normalized uniform superposition of the $N-1$ unmarked inputs, which is orthogonal to $|w\rangle$ and lies in the same plane. Then

$$|s\rangle = \sin\theta\,|w\rangle + \cos\theta\,|w_\perp\rangle\,, \qquad \sin\theta = \frac{1}{\sqrt{N}}, \quad 0 < \theta \leqslant \frac{\pi}{2}. \tag{1.3}$$

Explicitly, with one marked item $w$,

$$|w_\perp\rangle = \frac{1}{\sqrt{N-1}} \sum_{x\neq w} |x\rangle\,, \qquad \langle w|w_\perp\rangle = 0, \qquad \langle w_\perp|w_\perp\rangle = 1. \tag{1.4}$$

Splitting the sum in (1.2) into the term $x = w$ and the rest gives $|s\rangle = \frac{1}{\sqrt{N}}\,|w\rangle + \sqrt{\frac{N-1}{N}}\,|w_\perp\rangle$, which is (1.3) with $\sin\theta = 1/\sqrt{N}$ and $\cos\theta = \sqrt{(N-1)/N}$. For large $N$ the angle $\theta$ is small, so $|s\rangle$ is nearly parallel to $|w_\perp\rangle$ and nearly orthogonal to the target $|w\rangle$. The task is to rotate the state within this plane from $|s\rangle$ toward $|w\rangle$.

The oracle already performs one of the two operations needed. On the plane, $O_f = \mathbb{I} - 2\,|w\rangle\langle w|$ fixes $|w_\perp\rangle$ and negates $|w\rangle$: it is the reflection about the line spanned by $|w_\perp\rangle$, or equivalently about the hyperplane orthogonal to $|w\rangle$. A single reflection about $|w_\perp\rangle$ cannot bring the state closer to $|w\rangle$, since it preserves $|\langle w|\psi\rangle|$, as the failed attempt of the previous section showed. Two reflections, about two different lines, compose to a rotation. The second line must be one we can reflect about without knowing $w$, and the natural choice is the line spanned by $|s\rangle$, which we know how to prepare.

### 1.2.1 Inversion about the mean

**Definition 1.3: Diffusion operator**

The diffusion operator on $n$ qubits is

$$D = 2\,|s\rangle\langle s| - \mathbb{I} = H^{\otimes n}\,(2\,|0\rangle\langle 0| - \mathbb{I})\,H^{\otimes n}, \tag{1.5}$$

where $|0\rangle$ stands for $|0\rangle^{\otimes n}$. It fixes $|s\rangle$ and negates every state orthogonal to $|s\rangle$, so it is the reflection about the line spanned by $|s\rangle$. It does not depend on $f$ and makes no query.

The second expression, verified below, shows that $D$ is a circuit of $2n$ Hadamard gates around a single operation that flips the sign of every computational basis state other than $|0\rangle^{\otimes n}$, or equivalently, up to a global phase, of $|0\rangle^{\otimes n}$ alone. Acting on an amplitude vector, $D$ replaces each amplitude by its reflection through the mean of all amplitudes, which is why it is also called inversion about the mean. Since $O_f$ has pushed the marked amplitude below the mean and every other amplitude slightly above it, inversion about the mean lifts the marked amplitude well above the rest. One Grover iteration is the product $G = D\,O_f$: the oracle first, then the diffusion.

Let $|\psi\rangle = \sum_x a_x |x\rangle$ be an arbitrary state and let $\bar{a} = \frac{1}{N}\sum_x a_x$ be the mean amplitude. Since $\langle s|\psi\rangle = \frac{1}{\sqrt{N}}\sum_x a_x = \sqrt{N}\,\bar{a}$,

$$D|\psi\rangle = 2\sqrt{N}\,\bar{a}\,|s\rangle - |\psi\rangle = \sum_x \big(2\bar{a} - a_x\big)|x\rangle\,. \tag{1.6}$$

Each amplitude is replaced by its mirror image through the mean: an amplitude at distance $d$ below the mean moves to distance $d$ above it. The mean itself is unchanged. This is the description of $D$ used in Example 1.1. After the oracle acts on $|s\rangle$ with one marked item, the marked amplitude is $-1/\sqrt{N}$ and the $N-1$ others are $+1/\sqrt{N}$, so the mean is $\bar{a} = (N-2)/(N\sqrt{N})$, slightly below $1/\sqrt{N}$. Inversion sends the unmarked amplitudes to $2\bar{a} - 1/\sqrt{N} = (N-4)/(N\sqrt{N})$, slightly smaller, and the marked amplitude to $2\bar{a} + 1/\sqrt{N} = (3N-4)/(N\sqrt{N})$, approximately three times its original size. This agrees with $\sin 3\theta \approx 3\sin\theta$ for small $\theta$; compare Theorem 1.1 below.

The identity $D = H^{\otimes n}(2|0\rangle\langle 0| - \mathbb{I})H^{\otimes n}$ follows from $H^{\otimes n}|0\rangle^{\otimes n} = |s\rangle$ and $H^{\otimes n}H^{\otimes n} = \mathbb{I}$: conjugating $2|0\rangle\langle 0| - \mathbb{I}$ by $H^{\otimes n}$ replaces $|0\rangle^{\otimes n}$ by $|s\rangle$ and leaves $\mathbb{I}$ unchanged. The operator $2|0\rangle\langle 0| - \mathbb{I}$ fixes $|0\rangle^{\otimes n}$ and negates every other computational basis state. Up to an unobservable global phase of $-1$ it is the same as the operator that negates $|0\rangle^{\otimes n}$ alone, which can be built from $X$ gates on every qubit, a $Z$ gate on the last qubit controlled by all the others, and $X$ gates on every qubit again.

### 1.2.2 Rotation in the invariant plane

Both $|w\rangle$ and $|w_\perp\rangle$ are eigenvectors of $O_f = \mathbb{I} - 2|w\rangle\langle w|$, with eigenvalues $-1$ and $+1$, so $O_f$ maps the plane $\mathcal{P} = \mathrm{span}\{|w\rangle, |w_\perp\rangle\}$ to itself. The operator $D = 2|s\rangle\langle s| - \mathbb{I}$ maps any $|\psi\rangle$ to $2\langle s|\psi\rangle|s\rangle - |\psi\rangle$, which lies in $\mathcal{P}$ whenever $|\psi\rangle$ does, because $|s\rangle \in \mathcal{P}$. Hence $G = D\,O_f$ preserves $\mathcal{P}$, and since the initial state $|s\rangle$ lies in $\mathcal{P}$, the entire computation takes place there. Both $O_f$ and $D$ have real matrix entries in the basis $\{|w\rangle, |w_\perp\rangle\}$, so the state remains a real unit vector in $\mathcal{P}$ and can be described by a single angle.

**Theorem 1.1** Rotation by $2\theta$

Let $\theta$ be defined by (1.3). The operator $G = D\,O_f$ maps the plane spanned by $|w\rangle$ and $|w_\perp\rangle$ to itself and acts on it as the rotation by the angle $2\theta$ in the direction from $|w_\perp\rangle$ toward $|w\rangle$. Consequently, after $k$ iterations applied to $|s\rangle$,

$$G^k|s\rangle = \sin\big((2k+1)\theta\big)|w\rangle + \cos\big((2k+1)\theta\big)|w_\perp\rangle\,, \tag{1.7}$$

and a computational-basis measurement then yields $w$ with probability $P_k = \sin^2\big((2k+1)\theta\big)$.

***Proof:*** Write a general unit vector in the plane as $|\psi_\varphi\rangle = \sin\varphi\,|w\rangle + \cos\varphi\,|w_\perp\rangle$, so that $|s\rangle = |\psi_\theta\rangle$. The oracle negates the $|w\rangle$ component and fixes the $|w_\perp\rangle$ component, so $O_f|\psi_\varphi\rangle = |\psi_{-\varphi}\rangle$. The diffusion operator is the reflection about $|\psi_\theta\rangle$, and a reflection about the line at angle $\theta$ sends the line at angle $\varphi$ to the line at angle $2\theta - \varphi$, so $D|\psi_\varphi\rangle = |\psi_{2\theta-\varphi}\rangle$; the trigonometric verification follows the proof. Composing, $G|\psi_\varphi\rangle = |\psi_{2\theta+\varphi}\rangle$, which is the rotation by $2\theta$. Starting from $\varphi = \theta$ and iterating $k$ times gives $\varphi = (2k+1)\theta$, which is (1.7), and the Born rule gives $P_k$. ■

Explicitly, for $|\psi_\varphi\rangle = \sin\varphi\,|w\rangle + \cos\varphi\,|w_\perp\rangle$ we have $\langle s|\psi_\varphi\rangle = \sin\theta\sin\varphi + \cos\theta\cos\varphi = \cos(\theta-\varphi)$, so

$$\begin{aligned} D|\psi_\varphi\rangle &= 2\cos(\theta-\varphi)\big(\sin\theta\,|w\rangle + \cos\theta\,|w_\perp\rangle\big) - \sin\varphi\,|w\rangle - \cos\varphi\,|w_\perp\rangle && (1.8)\\ &= \big(\sin(2\theta-\varphi) + \sin\varphi - \sin\varphi\big)|w\rangle + \big(\cos(2\theta-\varphi) + \cos\varphi - \cos\varphi\big)|w_\perp\rangle = |\psi_{2\theta-\varphi}\rangle\,, && (1.9)\end{aligned}$$

using $2\cos A\sin B = \sin(A+B) - \sin(A-B)$ and $2\cos A\cos B = \cos(A+B) + \cos(A-B)$ with $A = \theta - \varphi$ and $B = \theta$. Together with $O_f\,|\psi_\varphi\rangle = |\psi_{-\varphi}\rangle$ this gives $G\,|\psi_\varphi\rangle = |\psi_{\varphi+2\theta}\rangle$. In matrix form, in the ordered basis $\{|w\rangle, |w_\perp\rangle\}$,

$$O_f = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}, \qquad G = D\,O_f = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix}, \tag{1.10}$$

where the entries of $D$ follow from $2\sin^2\theta - 1 = -\cos 2\theta$ and $2\sin\theta\cos\theta = \sin 2\theta$. Each of $O_f$ and $D$ has determinant $-1$, as a reflection must, and their product has determinant $+1$ and is the rotation matrix by $2\theta$ acting on the coordinate vector $(\sin\varphi, \cos\varphi)$.

The picture to carry is a clock hand. It starts at the small angle $\theta$ above the $|w_\perp\rangle$ axis, and each iteration advances it by $2\theta$ toward the $|w\rangle$ axis at angle $\pi/2$. The success probability is the square of the height of the hand. Each iteration costs one query, and the hand advances by the same angle every time, regardless of how far it has already turned.

### 1.2.3 The four-item case

**Example 1.1** (One marked item among four)

Take $n = 2$, so $N = 4$, and let the marked item be $w = 10$. Then $\sin\theta = 1/2$, so $\theta = \pi/6$, and one iteration turns the hand from $\pi/6$ to $3\pi/6 = \pi/2$, exactly onto $|w\rangle$. In amplitudes: the four bars of $|s\rangle$ all have height $\frac{1}{2}$. The oracle flips the bar at 10 to $-\frac{1}{2}$ and leaves the other three at $\frac{1}{2}$, so the mean of the four amplitudes drops to $\frac{1}{4}$. Inversion about the mean sends each amplitude $a$ to $2\cdot\frac{1}{4} - a$: the three unmarked bars go from $\frac{1}{2}$ to 0 and the marked bar goes from $-\frac{1}{2}$ to 1.

| | $\lvert 00\rangle$ | $\lvert 01\rangle$ | $\lvert 10\rangle$ | $\lvert 11\rangle$ |
|---|---|---|---|---|
| initial state $\lvert s\rangle$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ |
| after $O_f$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $-\frac{1}{2}$ | $\frac{1}{2}$ |
| after $D$ | 0 | 0 | 1 | 0 |

A single query finds the marked item with certainty, where a classical algorithm needs up to three queries and finds it in one only by luck.

The point of the example is not the number four but the mechanism: the oracle turns the marked amplitude negative, and the diffusion operator, which knows nothing about $w$, converts that sign into a large positive amplitude by comparing every amplitude with the mean. Both operations act on all $N$ amplitudes at once, and neither is a lookup.

In vector notation the same computation reads as follows. Take $n = 2$, $N = 4$, and $w = 10$. In the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ the initial amplitude vector is $\frac{1}{2}(1,1,1,1)$. The oracle negates the third entry: $\frac{1}{2}(1,1,-1,1)$. The mean is $\bar{a} = \frac{1}{4}\cdot\frac{1}{2}(1+1-1+1) = \frac{1}{4}$. Inversion about the mean gives $2\bar{a} - a_x = \frac{1}{2} - a_x$, hence $(0,0,1,0)$. The state after one iteration is exactly $|10\rangle$, and measurement yields $w$ with probability 1.

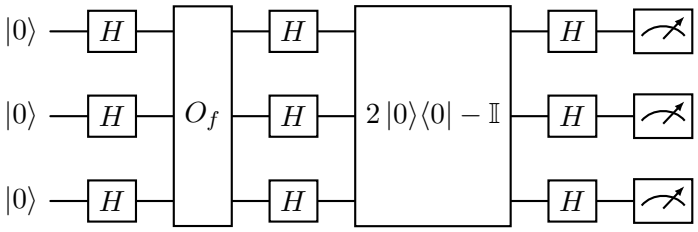
The angle picture gives the same result: $\sin\theta = \frac{1}{2}$, so $\theta = \pi/6$ and $3\theta = \pi/2$, and $P_1 = \sin^2(\pi/2) = 1$. A second iteration gives $5\theta = 5\pi/6$ and $P_2 = \sin^2(5\pi/6) = \frac{1}{4}$; explicitly, the oracle sends $(0,0,1,0)$ to $(0,0,-1,0)$, the mean is $-\frac{1}{4}$, and inversion gives $\frac{1}{2}(-1,-1,1,-1)$, whose marked entry has squared modulus $\frac{1}{4}$. A third iteration gives $7\theta = 7\pi/6$ and $P_3 = \frac{1}{4}$, a fourth gives $9\theta = 3\pi/2$ and $P_4 = 1$ again. The success probability cycles with period three in $k$ for this instance.

**Remark 1.1** Overshooting

The rotation does not stop when it reaches $|w\rangle$. If $(2k+1)\theta$ exceeds $\pi/2$, the hand moves past the $|w\rangle$ axis and its height begins to fall; at $(2k+1)\theta = \pi$ the state is back on the $|w_\perp\rangle$ axis and the success probability is zero. The success probability $\sin^2((2k+1)\theta)$ is periodic in $k$. In Example 1.1, a second iteration turns the hand to $5\pi/6$ and the probability of finding 10 falls from 1 to $\frac{1}{4}$, no better than a random guess. Grover's algorithm is not a procedure that improves monotonically the longer it runs. The number of iterations is a parameter that must be chosen, and choosing it requires knowing $\theta$, hence $N$ and the number of solutions.

### 1.2.4 The iteration circuit

The following circuit, drawn for $n = 3$, prepares $|s\rangle$ and applies one iteration $G = D\,O_f$. The box marked $O_f$ is the phase oracle, obtained from $U_f$ by the kickback construction of **??** with the ancilla omitted from the drawing. The box marked $2\,|0\rangle\langle 0| - \mathbb{I}$ flips the sign of every computational basis state other than $|000\rangle$. The full algorithm repeats the portion between the first column of Hadamard gates and the measurements $k_{\text{opt}}$ times, where the optimal number of iterations $k_{\text{opt}}$ is determined in section 1.3.1.




Reading left to right, the circuit computes $D\,O_f\,H^{\otimes 3}\,|000\rangle = G\,|s\rangle$, since $H^{\otimes 3}(2\,|0\rangle\langle 0| - \mathbb{I})H^{\otimes 3} = D$. The three middle gates form the diffusion operator; only the $O_f$ box depends on $f$, and only it counts as a query. The Hadamard gates at the end of one iteration and the beginning of the diffusion in the next do not cancel, because the oracle sits between them.

## 1.3 Query complexity

### 1.3.1 The optimal number of iterations

The success probability is largest when $(2k+1)\theta$ is closest to $\pi/2$, that is, when $k$ is closest to $\pi/(4\theta) - \frac{1}{2}$. For large $N$ we have $\theta \approx \sin\theta = 1/\sqrt{N}$, so the optimal number of iterations is

$$k_{\text{opt}} \approx \frac{\pi}{4}\sqrt{N}, \tag{1.11}$$

and with this choice the final angle is within $\theta$ of $\pi/2$, so the failure probability is at most $\sin^2\theta = 1/N$. Each iteration makes one query, so Grover's algorithm finds the marked item with about $\frac{\pi}{4}\sqrt{N}$ queries, against about $N/2$ for the best classical strategy. This is a quadratic improvement in the number of queries: the exponent is halved. Since $N = 2^n$, the quantum query count is about $2^{n/2}$, which is still exponential in the number of bits $n$.

The estimate can be made precise, and it is convenient to do so for a general number $M$ of marked items, for which $\sin\theta = \sqrt{M/N}$, as section 1.3.2 shows; the case treated so far is $M = 1$. From (1.7), $P_k = \sin^2((2k+1)\theta)$. This equals 1 exactly when $(2k+1)\theta = \pi/2$, that is, when $k = \pi/(4\theta) - \frac{1}{2}$ is an integer, which happens for $N = 4$ and rarely otherwise. In general let $k_{\text{opt}}$ be the integer nearest to $\pi/(4\theta) - \frac{1}{2}$. Then $|(2k_{\text{opt}}+1)\theta - \pi/2| \leqslant \theta$, so the final angle differs from $\pi/2$

by at most $\theta$ and

$$P_{k_{\text{opt}}} \geqslant \cos^2\theta = 1 - \sin^2\theta = 1 - \frac{M}{N}. \tag{1.12}$$

For large $N$, $\theta \approx \sin\theta = \sqrt{M/N}$, since $\sin\theta \leqslant \theta \leqslant \sin\theta/\cos\theta$, and therefore $k_{\text{opt}} \approx \frac{\pi}{4}\sqrt{N/M}$, which is (1.11) for $M = 1$. For $N = 2^{20}$ and $M = 1$ this is about 804 iterations, against an expected 524 288 classical queries. If the algorithm fails, which happens with probability at most $M/N$, one checks the measured $x$ with a single classical evaluation of $f$ and repeats; the expected total number of queries is therefore still of order $\sqrt{N/M}$.

### 1.3.2 Several marked items

When there are $M$ marked items, the role of $|w\rangle$ is taken by the normalized uniform superposition $|W\rangle$ of the marked inputs. The argument of Theorem 1.1 goes through unchanged with $\sin\theta = \sqrt{M/N}$, as verified below, and the optimal number of iterations becomes about $\frac{\pi}{4}\sqrt{N/M}$. More solutions means a larger initial angle and fewer iterations, as one expects. The comparison with the classical count of about $N/M$ is again quadratic. Since the optimal count depends on $M$, the number of solutions must be known or estimated; Remark 1.2 discusses what can be done when it is unknown.

Let $W$ be the set of marked items with $|W| = M$, and define the normalized states

$$|W\rangle = \frac{1}{\sqrt{M}} \sum_{x \in W} |x\rangle\,, \qquad |W_\perp\rangle = \frac{1}{\sqrt{N-M}} \sum_{x \notin W} |x\rangle\,. \tag{1.13}$$

Then $|s\rangle = \sqrt{\frac{M}{N}}\,|W\rangle + \sqrt{\frac{N-M}{N}}\,|W_\perp\rangle$, so $\sin\theta = \sqrt{M/N}$. The phase oracle acts as $-1$ on $|W\rangle$ and as $+1$ on $|W_\perp\rangle$, because every term of $|W\rangle$ is marked and no term of $|W_\perp\rangle$ is. The argument of section 1.2.2 applies verbatim with $|w\rangle$ replaced by $|W\rangle$, and after $k$ iterations the state is $\sin((2k+1)\theta)\,|W\rangle + \cos((2k+1)\theta)\,|W_\perp\rangle$. Measuring it yields some marked item with probability $\sin^2((2k+1)\theta)$, each marked item being equally likely. The optimal iteration count is about $\frac{\pi}{4}\sqrt{N/M}$. If $M = N/4$ then $\theta = \pi/6$ and one iteration succeeds with certainty, generalizing Example 1.1; if $M > N/2$ then $\theta > \pi/4$ and already the first iteration overshoots, but in that case a random guess succeeds with probability above $\frac{1}{2}$ and no quantum algorithm is needed.

> **Remark 1.2** Unknown number of solutions
>
> The optimal iteration count depends on $M$ through $\theta$, and Remark 1.1 showed that a wrong count can be catastrophic: the count that is optimal for $M = 1$ overshoots badly when $M = 4$, since the rotation angle has doubled. If $M$ is unknown, one remedy is to run the algorithm with randomly chosen iteration counts drawn from a range that is multiplied by a constant factor between 1 and 4/3 after each failure; this finds a solution after an expected number of queries still proportional to $\sqrt{N/M}$, with a worse constant [2]. Another is to estimate $M$ first by a quantum counting procedure built on the phase estimation technique of **??**; the counting procedure itself is outside this course. Either way, the naive picture of running Grover's algorithm for a fixed number of steps on an arbitrary $f$ is incomplete.

### 1.3.3 The lower bound

Could a cleverer algorithm do better than $\sqrt{N}$? Within the model of Definition 1.1, where the only access to $f$ is through queries to $O_f$, the answer is no.

**Theorem 1.2** Lower bound for unstructured search

Any quantum algorithm that, for every marking function $f$ with exactly one marked item among $N$, finds the marked item with probability at least $\frac{2}{3}$ using only queries to $O_f$, must make $\Omega(\sqrt{N})$ queries, meaning at least a constant multiple of $\sqrt{N}$. Consequently Grover's algorithm is asymptotically optimal among quantum algorithms for unstructured search.

The theorem is due to Bennett, Bernstein, Brassard, and Vazirani [3], and it predates Grover's algorithm; we state it without proof. The idea behind Theorem 1.2 is that a state which has been queried $k$ times can differ from the state produced without any marked item by an amount of order $k/\sqrt{N}$ on average over the choice of $w$, so distinguishing the $N$ possible positions of $w$ requires $k$ of order $\sqrt{N}$. Later work sharpened the bound to show that even the constant $\pi/4$ in (1.11) cannot be improved [4]. Two features of the statement deserve attention. It is a statement about queries, not about total gate count or running time. And it is a statement about the unstructured model, in which the algorithm knows nothing about $f$ beyond what its queries return. Both features matter for what follows.

**Wrong Concept 1.2: Grover search gives an exponential speedup**

It does not. Grover's algorithm reduces about $N$ queries to about $\sqrt{N}$, and Theorem 1.2 shows that no quantum algorithm in the same model does better. For a search over $n$-bit strings this is a reduction from $2^n$ to $2^{n/2}$, which is a halving of the exponent, not a removal of it. Doubling the number of bits still squares the work. Exponential speedups, such as the one claimed for factoring later in the course, come from problems with algebraic structure that a quantum algorithm exploits. Unstructured search has no such structure, by definition, and its speedup is correspondingly modest.

### 1.3.4 Limits of the speedup

The query count is a clean measure but it is not the cost of running the algorithm. Three further costs must be considered before a claimed advantage becomes real.

First, the oracle must be built. The algorithm requires $O_f$ as a reversible quantum circuit, and each iteration executes that circuit once. The case of searching a cipher key space, and why parallel hardware weakens the advantage further, is taken up in **??**. If instead $f$ is a lookup in a stored table of $N$ entries, then building an oracle that can be queried in superposition requires the table to be loaded into a quantum-accessible form, which costs at least $N$ operations and cancels the advantage for a one-off search; only when the same loaded table serves many searches is that cost amortized. Grover's algorithm searches for a preimage of a function we can compute, not for a record in a database we have stored.

Second, the input must be prepared. The uniform superposition costs only $n$ Hadamard gates, so preparation is cheap here. But the whole analysis assumes the algorithm starts from $|s\rangle$ and has no prior information about where $w$ lies. When prior information exists, it is a form of structure, and one should ask whether a classical algorithm can use it.

Third, and most important, the comparison is with classical algorithms that are also forbidden to use structure. Almost every search problem that arises in practice has structure. A sorted list is searched classically in $\log_2 N$ queries by bisection, which beats $\sqrt{N}$ by an enormous margin, and Grover's algorithm offers nothing in return. Constraint satisfaction problems are attacked classically by solvers that prune the search space using the constraints, and their running time on realistic instances is far below $N$; a quantum algorithm that ignores the constraints and runs Grover's

algorithm on the $2^n$ assignments is competing against the wrong baseline. Grover's algorithm can sometimes be combined with structured classical methods to gain a square root on part of the work, but it does not by itself turn a hard optimization problem into an easy one.

The comparison is fair only when both sides are restricted to queries. The moment $f$ has exploitable structure, the classical side is no longer bound by $N/M$. If the inputs are sorted so that $f$ is monotone, bisection finds the boundary in $\log_2 N$ queries. If $f$ is a polynomial identity, algebra finds the roots directly. If $f$ encodes a satisfiability instance, backtracking with constraint propagation rules out large regions of the search space without querying them individually. In each case the classical algorithm uses knowledge about $f$ that the query model forbids, and the quantum algorithm, which uses only $O_f$, cannot match it. The advantage of Grover's algorithm is therefore most reliable precisely for functions that are as featureless as a cipher is designed to be, which is why its most concrete application is the attack on symmetric keys discussed in **??**, and why even there the advantage is a halving of the effective key length rather than a break.

**Remark 1.3** Scope of Grover's algorithm

Grover's algorithm is a provably optimal quantum procedure for a precisely defined problem: find a marked item using only queries to a recognizer. Its advantage is quadratic in the number of queries, requires the number of solutions to be known or estimated, and is measured against a classical baseline that is equally deprived of structure. It is not a general-purpose accelerator for optimization, it does not make problems in the class NP (a complexity class we name but do not define; complexity theory is outside this course) tractable, and it does not evaluate all inputs and select the best one. Its lasting importance is the mechanism: amplitude amplification by alternating reflections is a subroutine that appears inside many other quantum algorithms, where it improves a success probability $p$ to near certainty in about $1/\sqrt{p}$ repetitions instead of the classical $1/p$.

## 1.4 Summary

Unstructured search asks for a solution given only a recognizer for solutions, and the cost is counted in queries. The phase oracle from **??** marks a solution with a sign, which is a relative phase and is invisible to a direct measurement. The uniform superposition is not a search algorithm: measuring it returns a random candidate. The Grover iteration composes two reflections, the oracle about the hyperplane orthogonal to the marked state and the diffusion operator about the uniform state, and two reflections make a rotation by $2\theta$ with $\sin\theta = \sqrt{M/N}$ in the plane spanned by the two states. After $k$ iterations the success probability is $\sin^2((2k+1)\theta)$, which is maximized near $k \approx \frac{\pi}{4}\sqrt{N/M}$ and then decreases again, so the number of iterations must be chosen and the number of solutions must be known or estimated. The quadratic query advantage is optimal in the unstructured model, and it is quadratic, not exponential. Whether it translates into a practical advantage depends on the cost of the oracle circuit, on the impossibility of parallelizing the iterations, and above all on whether the problem really lacks structure that a classical algorithm could use.

Homework 3, part A, can be attempted after this lecture.

# References


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